

Final Year Project Document

Multi-Agent Market Research Platform

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Chapter 1

Introduction

The out-of-home advertising industry reports $9.46 billion in revenue, yet market researchers rely on fragmented data tools. Field workers record GPS coordinates, photographs, and surveys across separate applications. This disconnected approach demands roughly 50 staff-hours per 100 locations for manual data reconciliation. Analysts struggle to verify unmatched field evidence against digital campaign logs.

The proposed system provides a unified web and Android architecture to eliminate these bottlenecks. Administrators structure campaigns through a central web portal. Field staff capture observations via the Android application. A local SQLite database secures offline submissions and forces incremental synchronization to PostgreSQL servers when internet access returns.

The software deploys specialized artificial intelligence agents to automate analytical workflows. Distinct modules coordinate research planning, secondary data extraction, and cross-source verification. LangGraph-driven tools evaluate billboard image quality, process media attachments, and calculate performance metrics. The automated pipeline routes anomalous records directly to human reviewers for final validation.

Advertising agencies utilize this traceable workflow to publish reliable market intelligence. The unified pipeline protects data provenance from street-level capture to executive reports. Research firms eliminate repetitive manual validation and scale operations safely. The following sections describe the system purpose, system scope, development objectives, intended market, and target users.

1.1 Product Purpose

The proposed system is a centralized web and Android workspace for out-of-home advertising research. The unified platform organizes field surveys, GPS coordinates, and media attachments inside a secure PostgreSQL database. Older disconnected tools make organizations waste fifty staff-hours reconciling data for campaign locations. The Android application resolves these specific administrative bottlenecks through SQLite-based offline persistence and network synchronization. Specialized artificial intelligence agents analyze billboard image quality, cross-reference competitor data, and compile performance metrics. The automated workflow sends anomalous records directly to human reviewers for final validation. The new architecture replaces manual data entry tasks with verifiable audience measurement standards. Advertising agencies deploy this traceable

infrastructure to justify massive brand investments confidently.

1.2 Product Scope

Software engineering frameworks define the scope of the proposed system as the strict boundaries of development. The system scope identifies the exact deliverables within these limits. A clear scope establishes strict expectations for the software lifecycle. The approved documentation prevents scope creep during system construction. Development managers allocate time and personnel resources based on these defined constraints. The stakeholders utilize the finalized scope to lock exact delivery requirements. The official scope provides a rigid baseline for software validation. Quality assurance teams check the final application against this original baseline [1].

The following analytical modules outline the exact out-of-home advertising capabilities of the proposed system. These specific LangGraph and PostgreSQL features define what the automated market research platform delivers to agencies.

Module: User Management

- Company workspace creation and management.
- User registration and authentication.
- Role-based access and permissions.
- Device registration and authorization.
- Workflow-based information access.
- Account activity and audit records.

Module: Questionnaire Management

- Market-research questionnaire creation.
- Support for retail, billboard, competitor and field surveys.
- Question branching and response validation.
- English and Urdu questionnaire support.
- OTP and eligibility controls.
- Questionnaire version management.

Module: Campaign Management

- Multi-market campaign creation and management.
- Billboard and advertisement registration.
- Competitor campaign and advertisement records.
- Hierarchical location management.

- Field assignment and campaign scheduling.
- Campaign, location and evidence linking.

Module: Attendance Verification

- Field-agent location assignment.
- Geofence-based attendance verification.
- GPS-based visit validation.
- Visit timestamp recording.
- Location-based attendance records.
- Supporting visit evidence management.

Module: Data Synchronization

- SQLite-based offline data persistence.
- Incremental data synchronization.
- Submission queues and retry handling.
- Synchronization status tracking.
- Primary collection ID generation and management.
- Media association using collection IDs.

Module: Field Data-Collection

- Android-based field data collection.
- Assigned campaign and questionnaire access.
- Offline response and observation capture.
- GPS and timestamp collection.
- Photograph and audio capture.
- Secure local submission storage.

Module: AI Verification

- AI-based response quality checking.
- Evidence consistency verification.
- On-device preliminary AI processing.
- Real-time billboard and image analysis.
- Detection of missing or inconsistent information.
- Human review routing for uncertain results.

Module: Market Intelligence

- Secondary market-data collection.
- Competitor information collection.
- Web and document information extraction.
- Market and competitor comparison.
- Source and evidence tracking.
- Multi-market research analysis.

Module: Media Processing

- Image capture and management.
- Audio capture and management.
- Image quality assessment.
- Image enhancement where applicable.
- Audio quality and preprocessing.
- Media metadata and collection linking.

Module: Agent Orchestration

- Research planning agent coordination
- Data collection and extraction agent coordination.
- Verification and analysis agent coordination.
- Automated reporting and dashboard preparation.
- Coding Agent support for development tasks.
- Agent permissions, workflow control and human-review handling.

Module: Field Performance

- Field-agent activity tracking.
- Visit and task completion measurement.
- Evidence-completeness monitoring.
- Pitch recording and analysis.
- AI-assisted performance feedback.
- Performance scores and organization-defined rewards.

Module: AI-Agent Analytics

- Interactive market-data visualization.
- Campaign and location comparison.
- Market and competitor comparison.

- Automated AI-generated insights.
- Dashboard-based research reporting.
- Role-based analytics views.

1.3 Objectives

The software development methodologies define objectives as specific, measurable, achievable, relevant, and time-bound statements. These SMART statements translate high-level system goals into concrete technical outcomes. The clear objectives guide development priorities for the engineering team. The documented targets dictate proper resource allocation throughout the software lifecycle. The system stakeholders use these measurable statements to communicate expectations clearly. The finalized objectives provide a strict mathematical basis for system validation. The quality assurance evaluators compare the final software deliverables against these exact original criteria [2].

The proposed system delivers twelve functional modules by the final system evaluation. The User Management module establishes secure company workspaces and strict device authorization protocols. The Campaign Management module connects specific billboard advertisements to hierarchical market locations. The Field Data-Collection module enables offline questionnaire responses through the custom Android application. The Data Synchronization module pushes the SQLite records to the PostgreSQL server upon internet reconnection. The automatic synchronization tests guarantee zero duplicate submissions during these network transfers. The Attendance Verification module validates field surveyor visits through exact GPS records and geofences. The Media Processing module evaluates site photograph quality and preserves direct evidence links. The Market Intelligence module gathers secondary out-of-home advertising metrics for accurate competitor comparisons. The Agent Orchestration module manages the LangGraph artificial intelligence agents for research planning and data extraction. The automatic AI verification pipeline detects inconsistent survey information instantly. The routing algorithm directs uncertain field evidence directly to human reviewers. The Field Performance module evaluates visit completion rates and audio pitch recordings. The AI-Agent Analytics module presents complex campaign comparisons through interactive dashboards. The final evaluation compares manual and multi-agent workflows through strict completion time, accuracy, and failure recovery metrics.

The successful completion of these objectives will fulfill the core purpose of the proposed system. The unified architecture will connect fragmented research activities with traceable field evidence. The out-of-home advertising agencies will gain a common workspace for reliable campaign management and market comparison. The field staff will retain their survey submissions securely during sudden connectivity interruptions. The robust synchronization protocols will prevent data loss after network reconnection. The

data analysts will inspect supporting sources and reproduce numerical results effortlessly. The company managers will monitor field performance and compare campaign findings through automated dashboards. The final evaluation phase will document the specific strengths and limitations of the platform under repeatable tests. These measurable outcomes will demonstrate the practical connection between the software capabilities and the target market requirements.

1.4 Intended Market of Product

The system serves more than one market segment. The target customers include research companies, advertising agencies and businesses that study their own customers. The following subsections explain the needs of each market and how the system supports its research work.

Market Research Companies: The market research segment encompasses small consultancies and massive enterprise firms. These organizations collect survey responses and supervise field teams across multiple cities. The recent industry reports indicate massive data quality problems across this sector. Poor data quality costs these organizations millions of dollars annually. The manual reconciliation process requires fifty staff-hours for every one hundred campaign locations. The analysts struggle to keep GPS coordinates, media files, and survey responses consistent throughout a study. The proposed system will eliminate these exact administrative bottlenecks through secure company workspaces and offline SQLite collection. A firm will keep campaign responses and supporting evidence together automatically during fieldwork. The LangGraph agents will inspect source links instantly. The automated verification pipeline will route uncertain records to human reviewers. The field managers will monitor assignment completion through interactive dashboards. The firm will spend significantly less time repeating validation tasks. This streamlined workflow will help the firm manage research costs effectively. The researchers will present findings to clients with undeniable evidence.

Advertising and Media Agencies: The advertising agencies plan massive out-of-home campaigns for corporate clients. The out-of-home advertising sector generates over nine billion dollars in annual revenue. The digital screens represent a massive portion of that spending. These agencies need reliable metrics to justify these massive client investments. The current evidence arrives as disconnected photographs and paper survey files. This fragmented separation makes client reporting incredibly difficult. The executives require a reliable method to match field activity against the original campaign plan. The PostgreSQL database will connect specific digital advertisements with hierarchical market locations. An agency will utilize these database connections to validate field progress accurately. The managers will view surveyor attendance through geofenced dashboards before preparing the final campaign report. The direct evidence links will help the agency

explain performance metrics to executives. The agency will direct future field visits toward specific evidence gaps. The automated synchronization will prevent field teams from submitting duplicate records. The agencies will gain a massive competitive advantage through this verifiable audience measurement infrastructure.

Consumer Brands and Retail Businesses: The retail businesses conduct independent research to understand customer preferences. The target market includes local single-outlet shops and massive national brands. These businesses require accurate answers about competitor messaging and market differences. The internal staff members combine web survey files with physical field observations manually. The executives struggle to compare campaign results across different geographical locations. The manual data entry limits their ability to assess actual brand awareness. The proposed system will resolve these bottlenecks through configurable questionnaires and secondary market intelligence tools. A business will gather customer feedback reliably through the custom Android application. The company managers will compare competitor campaigns through the central web portal. The retail staff will spend less time matching spreadsheets together. The data analysts will have more time to examine actual customer feedback. The brand executives will utilize these verified findings to allocate future research budgets confidently. The organizations will deploy the artificial intelligence dashboards to track long-term consumer trends. The automated platform will enable smarter operational decisions for these retail chains.

1.5 Intended Users of the Product

The proposed system supports six distinct user groups across the market research lifecycle. Each professional group handles a different stage of the out-of-home advertising workflow. The following subsections define the specific administrative responsibilities of these users. The descriptions explain exactly how the new platform will assist each group.

Company Administrators: The company administrators control access to the central workspace. These technical users manage role permissions across fragmented software systems currently. The administrators struggle to secure data across disconnected field applications. The administrators will utilize the User Management module to create secure agency accounts. They will assign specific platform roles to managers, analysts, and field staff. The IT personnel will register authorized Android devices through the web portal. The system will restrict unauthorized mobile access automatically. The administrators will monitor account activity through immutable audit logs. These centralized controls will reduce repeated permission changes significantly. The security team will investigate access problems much faster. The agencies will protect their sensitive campaign data effectively.

Research Managers: The research managers translate client briefs into concrete

field assignments. These supervisors coordinate out-of-home advertising campaigns across multiple cities. The managers waste valuable hours gathering progress updates from separate field teams. The managers will use the Campaign Management module to assign specific billboard locations. They will configure complex branching rules within the Questionnaire Management module. The team leaders will coordinate the LangGraph artificial intelligence agents for automated data extraction. The managers will monitor field surveyor progress through real-time geofenced dashboards. The system will highlight missing evidence before the final reporting phase begins. This automated tracking will reduce the fifty staff-hours typically spent on manual coordination. The managers will deploy additional field resources toward specific evidence gaps efficiently.

Field Staff: The field staff visit assigned street locations to record billboard conditions. These surveyors ask audience questions and capture physical site photographs. Bad internet connections ruin their digital survey submissions. The field workers will use the custom Android application to view their daily campaign assignments. The surveyors will capture photographs, audio notes, and GPS coordinates offline. The SQLite database will store all pending submissions locally during internet outages. The Data Synchronization module will push these records to the PostgreSQL server upon network reconnection. The Attendance Verification module will validate their physical presence through strict geofences. The system will grade their task completion rates and pitch recordings automatically. The reliable offline architecture will prevent frustrating data loss completely.

Research Analysts: The research analysts evaluate collected survey responses and secondary market data. These professionals build client dashboards using fragmented spreadsheet files. The analysts struggle to match raw field photographs with the correct digital campaign logs. The data analysts will use the Market Intelligence module to compare competitor metrics accurately. The AI-Agent Analytics module will generate complex visualizations for out-of-home advertising campaigns. The LangGraph reporting agents will assist the analysts with heavy numerical calculations. The direct database links will allow analysts to trace every metric back to its original field photograph. The automated pipeline will calculate performance indicators against strict reference formulas. The analysts will access related survey records instantly. The agencies will deliver highly traceable market reports to their corporate clients.

Evidence Reviewers: The evidence reviewers examine anomalous records that fail automatic quality checks. These quality assurance workers validate missing or inconsistent field data manually. The reviewers waste extreme effort searching through separate media folders to find supporting photographs. The reviewers will evaluate uncertain survey submissions through the central web portal. The AI Verification pipeline will route questionable field records directly to these human workers. The Media Processing

module will display the original respondent answers alongside the site photographs. The reviewers will accept, correct, or exclude the flagged data according to strict agency procedures. The unified database will eliminate the need to search for missing media files manually. The quality assurance team will resolve data anomalies rapidly. The reviewers will guarantee absolute data accuracy for the final market analysis.

Survey Respondents: The survey respondents answer targeted questions about specific advertising campaigns. These everyday consumers participate through web links or physical street interviews. The poorly designed paper forms often confuse these participants with irrelevant questions. The respondents will complete digital questionnaires through the company website or surveyor devices. The branching logic will direct each consumer strictly to relevant campaign questions. The software will display forms in both English and Urdu languages. The instant validation prompts will point out incomplete answers immediately. The OTP eligibility controls will verify the authenticity of remote participants. The consumers will spend significantly less time reading unrelated survey sections. The clear digital interface will provide a much better experience for the general public. The system will capture accurate audience feedback effortlessly.

References

[1] I. Sommerville, "Software Engineering, 10th ed.," [Online]. Available: https://iansommerville.com/software-engineering-book/. [Accessed: Oct. 23, 2026].

[2] Project Management Institute, "A Guide to the Project Management Body of Knowledge (PMBOK Guide), 7th ed.," [Online]. Available: https://www.pmi.org/pmbok-guide-standards/foundational/pmbok. [Accessed: Oct. 23, 2026].